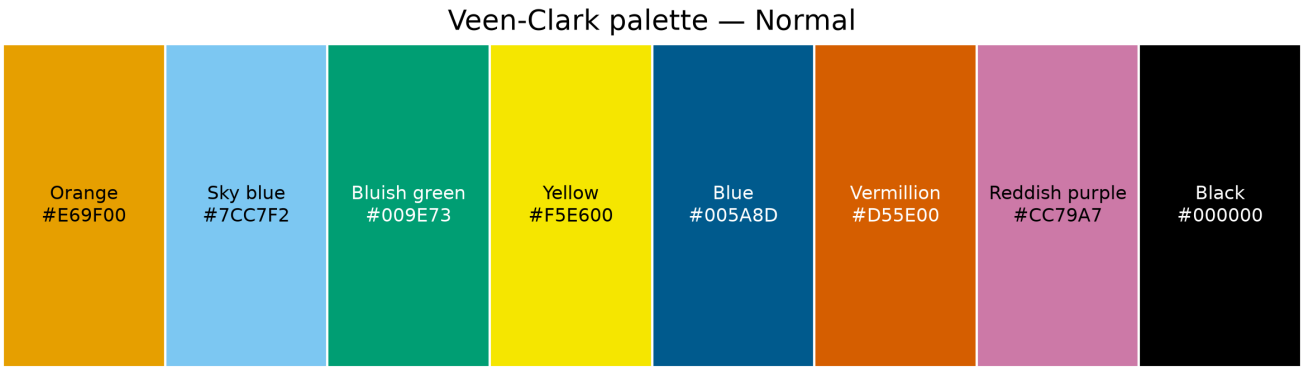
Veen-Clark palette

Version 0.1.0 evaluation

Archive DOI: 10.5281/zenodo.23418261



Veen-Clark palette in normal sRGB.

An Okabe-Ito derivative by Roland J. Veen and Charlie Clark. Sky blue is lightened and blue is darkened so the pair can represent related subgroups in scientific scatter and bar charts. The palette uses yellow #F5E600.

This report compares colour-difference metrics for the Veen-Clark and Okabe-Ito palettes.

Generated with PaletteBench 0.1.0 using Python 3.12.14. Exact inputs, hashes, dependencies, full-precision CSV/JSON data, and individual vector figures accompany this PDF in the repository.

Results

The intended sky-blue/blue subgroup pair has greater modelled separation relative to Okabe-Ito under every standard reported condition: normal, protan 100%, deutan 100%, tritan 100%, and grayscale.

Condition	Okabe-Ito	Veen-Clark	Change
Normal	22.31	37.92	+15.61
Protan 100%	20.75	36.03	+15.28
Deutan 100%	22.45	38.57	+16.12
Tritan 100%	19.95	35.08	+15.13
Grayscale	21.62	37.43	+15.81

The whole-palette result is mixed. The normal-vision 10th percentile rises by 2.73 Delta E00 and the protan-100% minimum rises by 3.56, but darkening blue reduces separation from black and produces the largest loss against vermillion under tritan 100% (-12.20 Delta E00).

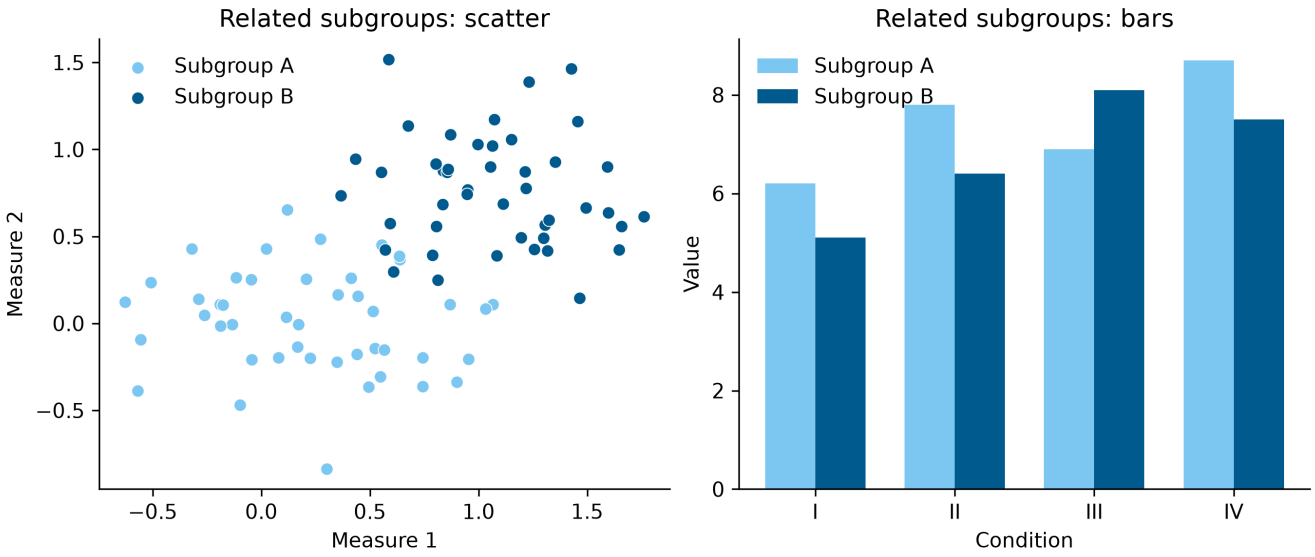


Figure 1. Synthetic data illustrating the two blues with circular markers and solid bar fills.

Yellow in context

Informal feedback from people with regular colour vision suggested that the original yellow could appear greenish and insufficiently distinct in presentations. This motivated the yellow choice; it was not a controlled participant study. The table reports modelled separation from orange and bluish green.

Condition	Pair	Okabe-Ito	Veen-Clark	Change
Normal	Yellow / orange	21.73	22.04	+0.31
Normal	Yellow / bluish green	38.04	39.94	+1.90
Protan 100%	Yellow / orange	15.17	15.81	+0.64
Protan 100%	Yellow / bluish green	28.05	28.93	+0.89
Deutan 100%	Yellow / orange	11.61	12.30	+0.70
Deutan 100%	Yellow / bluish green	33.90	35.47	+1.58
Tritan 100%	Yellow / orange	18.83	19.15	+0.32
Tritan 100%	Yellow / bluish green	43.57	43.25	-0.33
Grayscale	Yellow / orange	12.83	13.29	+0.47
Grayscale	Yellow / bluish green	23.32	23.78	+0.46

Yellow's normal-sRGB contrast ratio against white is 1.30:1. Pairwise colour separation does not establish visibility on a white slide. PaletteBench does not model projector colour errors or audience perception. Check the actual figure under its intended viewing conditions.

Baseline-relative comparison

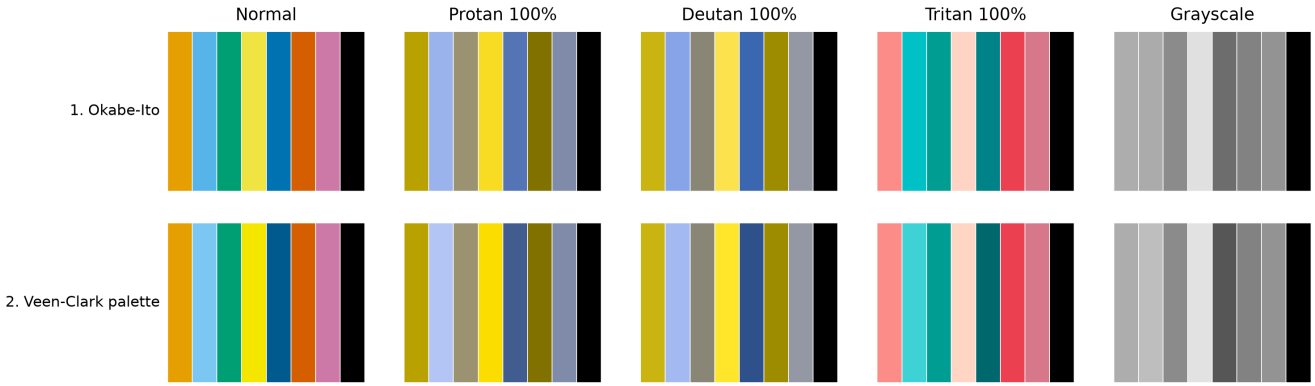


Figure 2. Okabe-Ito baseline and Veen-Clark under the standard reported conditions.

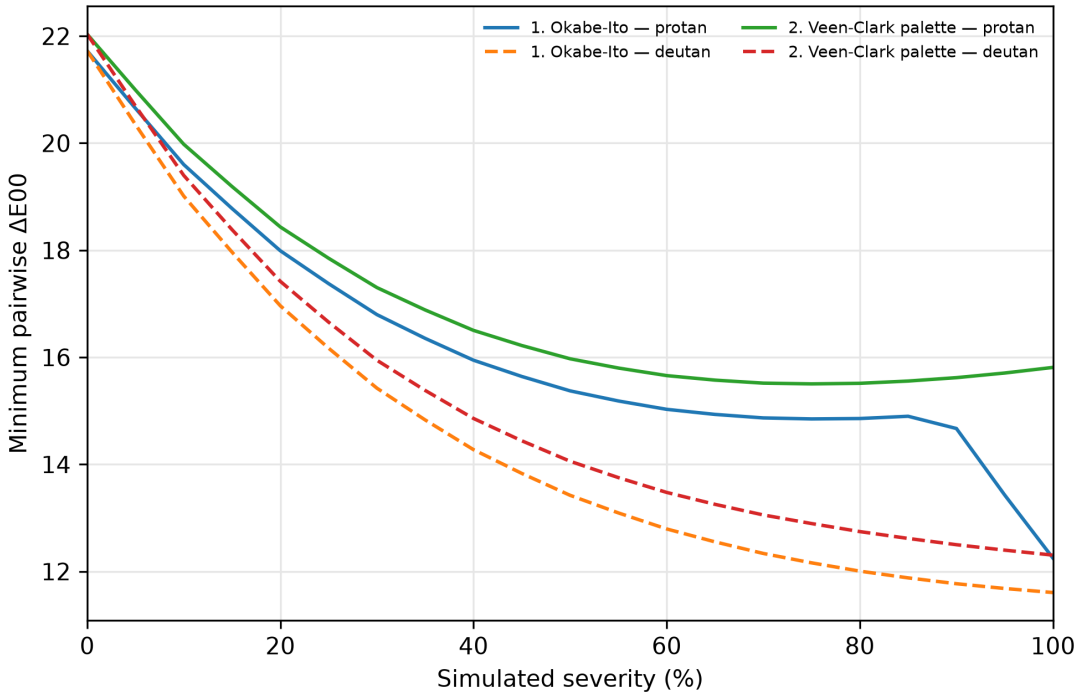


Figure 3. Whole-palette minimum Delta E00 by condition.

Baseline-relative comparison

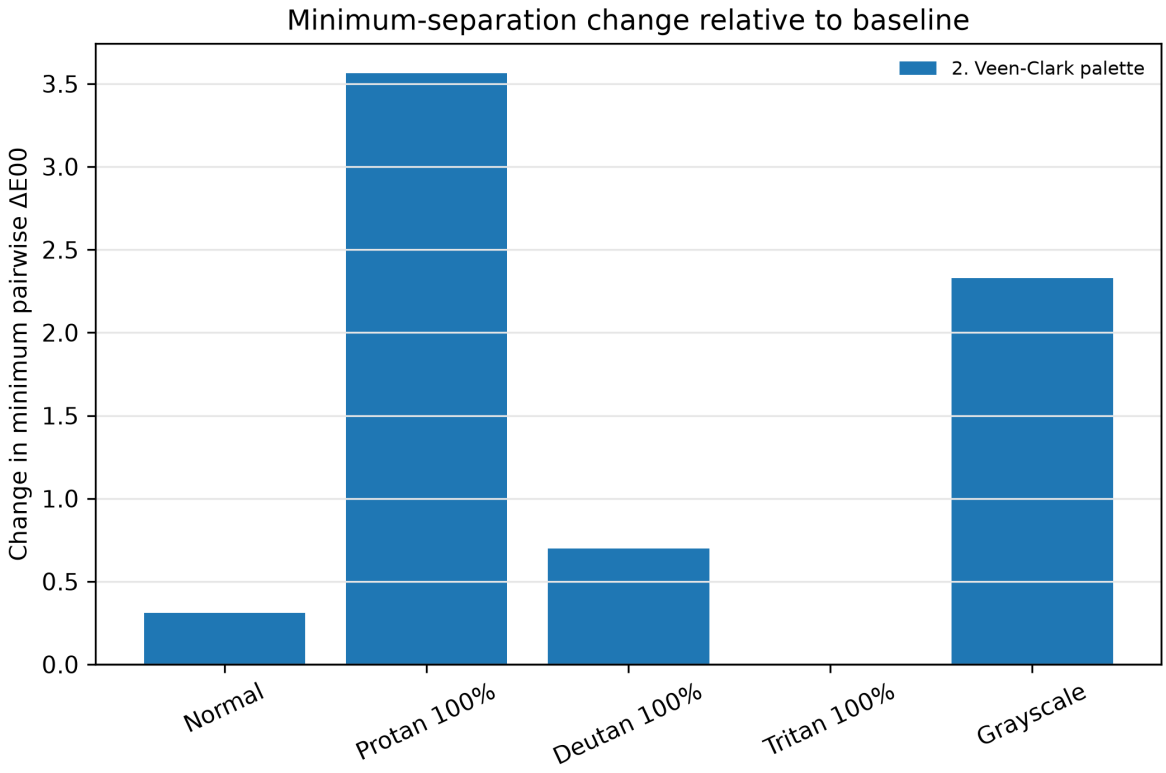


Figure 4. Change in minimum separation relative to Okabe-Ito.

Positive changes indicate greater modelled separation; negative changes indicate reduced modelled separation. Zero change means the minimum separation is unchanged.

Methods and assumptions

Six-digit values are interpreted as gamma-encoded sRGB. PaletteBench applies Colorspacious sRGB1+CVD simulations based on Machado, Oliveira and Fernandes, clips results to displayable sRGB, converts through CIE XYZ to CIE Lab under the sRGB D65 white point, and calculates CIEDE2000 colour differences with Colour Science for Python. Grayscale preserves WCAG relative luminance.

Severity percentages are simulation parameters, not clinical measurements. Partial tritan severity is omitted from the standard report. Clipping provenance and raw values are retained in the machine-readable audit directory.

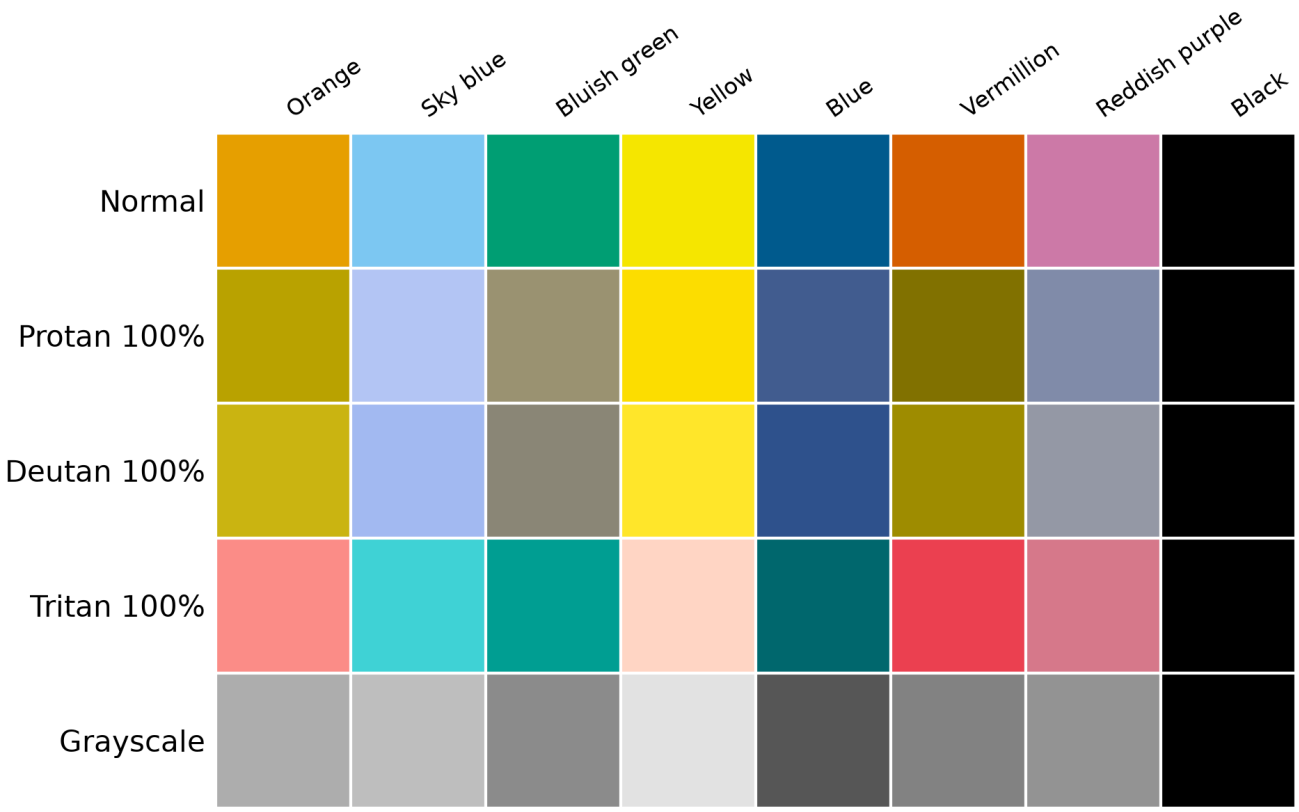


Figure 5. Normal, simulated CVD, and grayscale overview.

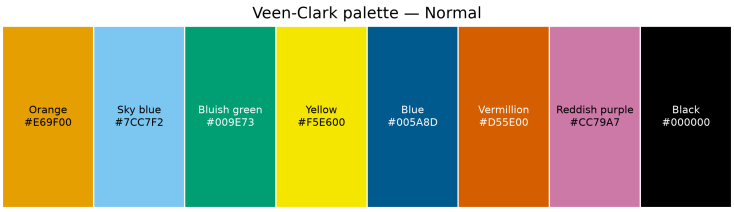
Palette characteristics

Colour	Hex	RGB	L*	Chroma	Luminance	Group
Orange	#E69F00	230 159 0	70.61	76.48	0.4162	-
Sky blue	#7CC7F2	124 199 242	77.01	30.81	0.5154	blue_family
Bluish green	#009E73	0 158 115	57.74	47.49	0.2569	-
Yellow	#F5E600	245 230 0	89.86	89.87	0.7601	-
Blue	#005A8D	0 90 141	36.43	34.62	0.0924	blue_family
Vermillion	#D55E00	213 94 0	54.19	76.67	0.2215	-
Reddish purple	#CC79A7	204 121 167	61.05	39.97	0.2930	-
Black	#000000	0 0 0	0.00	0.00	0.0000	-

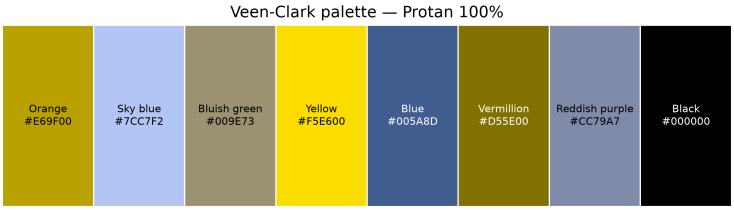
Condition-level summary

Condition	Min	Weakest pair	Mean	Median	10th pct.
normal	22.04	orange / yellow	49.68	49.65	33.71
protan20	18.43	orange / yellow	46.97	47.83	33.91
protan40	16.50	orange / yellow	45.05	44.26	28.67
protan60	15.66	orange / yellow	43.14	42.95	23.52
protan80	15.51	orange / yellow	41.88	42.61	19.46
protan100	15.81	orange / yellow	41.32	41.46	18.08
deutan20	17.42	orange / yellow	46.89	47.93	33.41
deutan40	14.86	orange / yellow	45.12	42.83	29.74
deutan60	13.48	orange / yellow	43.37	40.27	23.95
deutan80	12.74	orange / yellow	42.23	38.78	20.38
deutan100	12.30	orange / yellow	41.70	37.53	16.01
tritan100	10.87	orange / reddish_purple	44.53	52.05	15.79
grayscale	2.93	bluish_green / reddish_purple	26.21	20.88	5.79

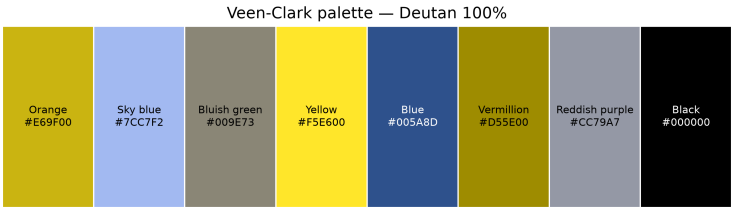
Palette strips



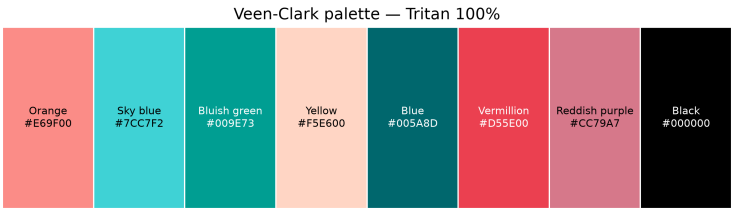
Normal sRGB



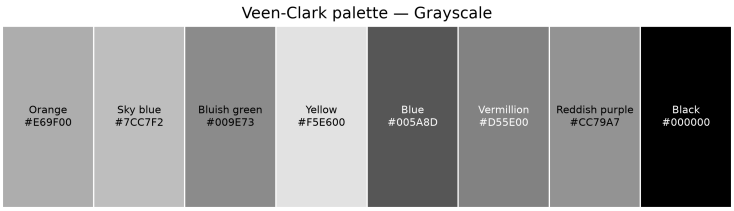
Simulated protan 100%



Simulated deutan 100%



Simulated tritan 100%



WCAG-luminance grayscale

Severity progressions

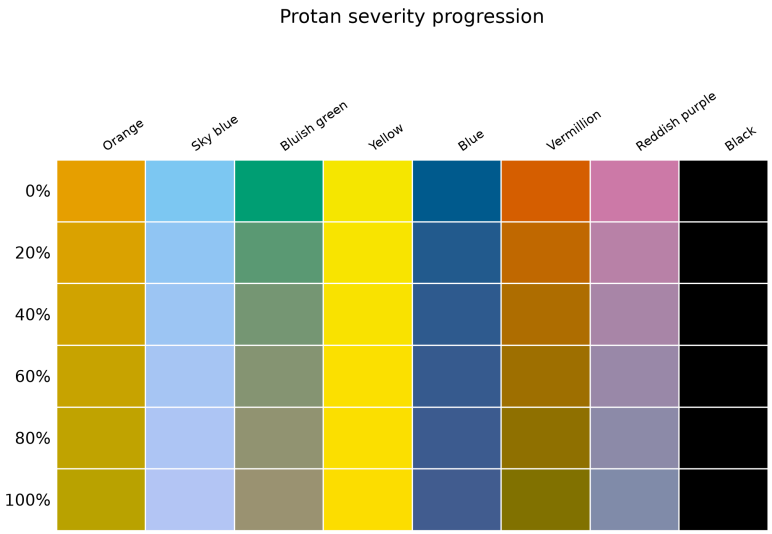


Figure 6. Protan simulation progression from 20% to 100%.

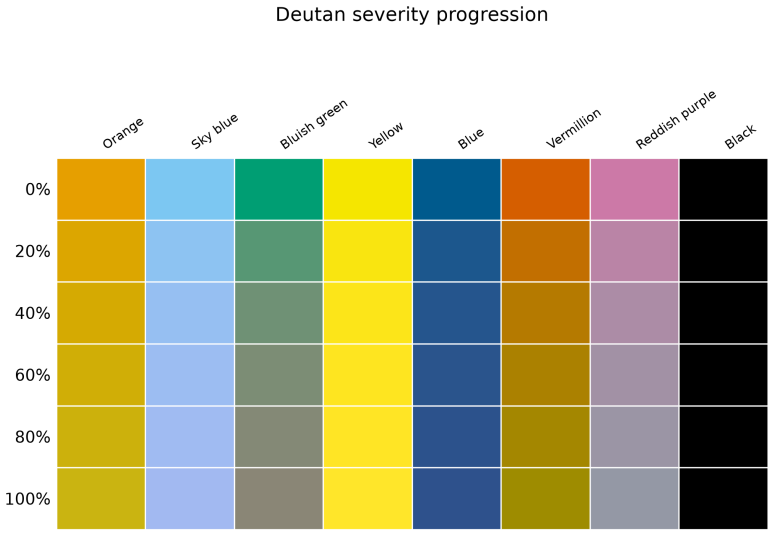


Figure 7. Deutan simulation progression from 20% to 100%.

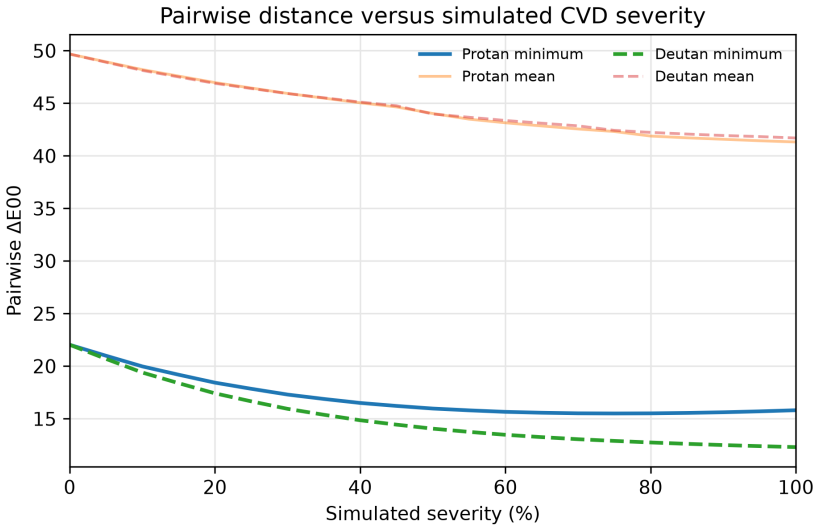
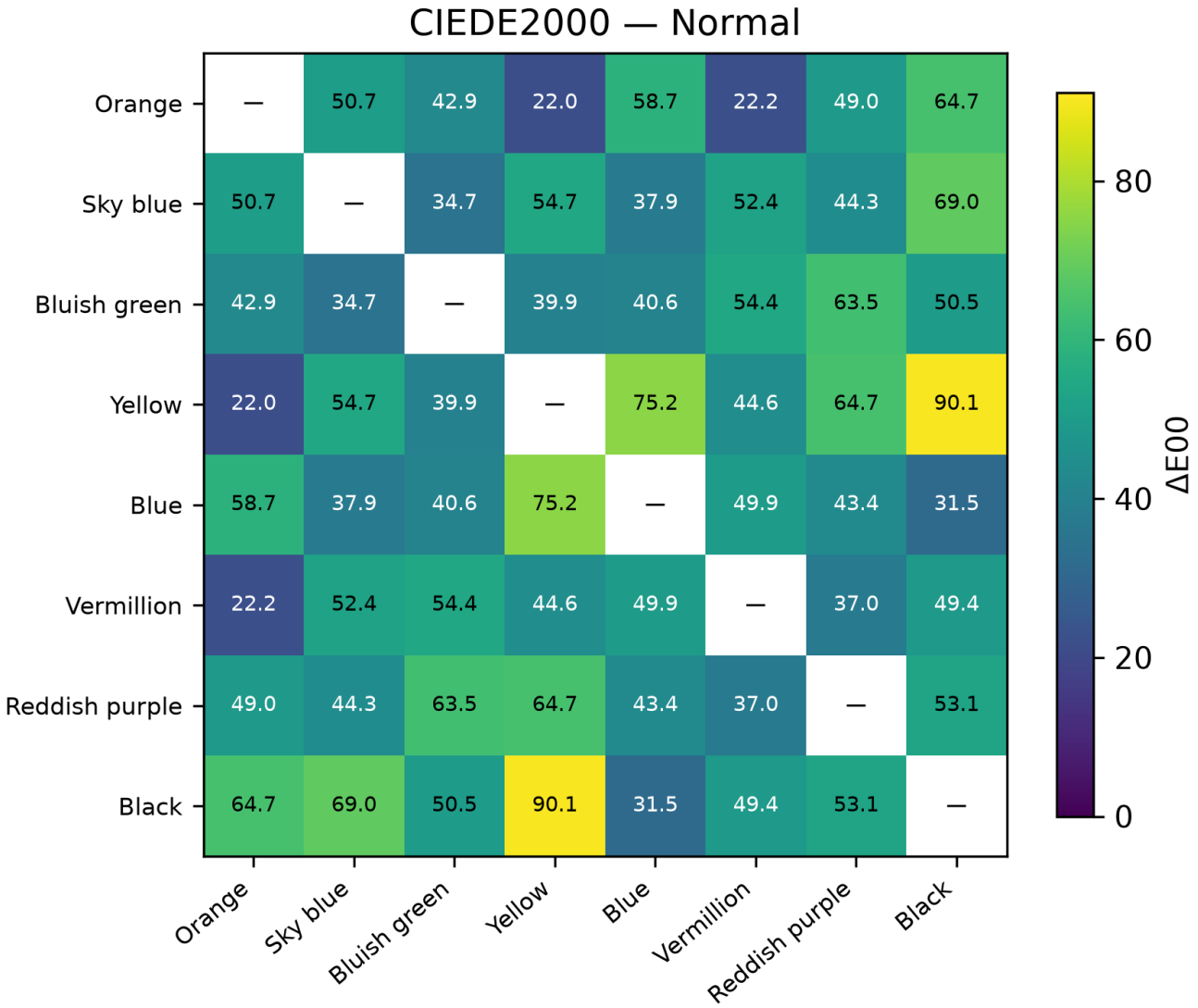


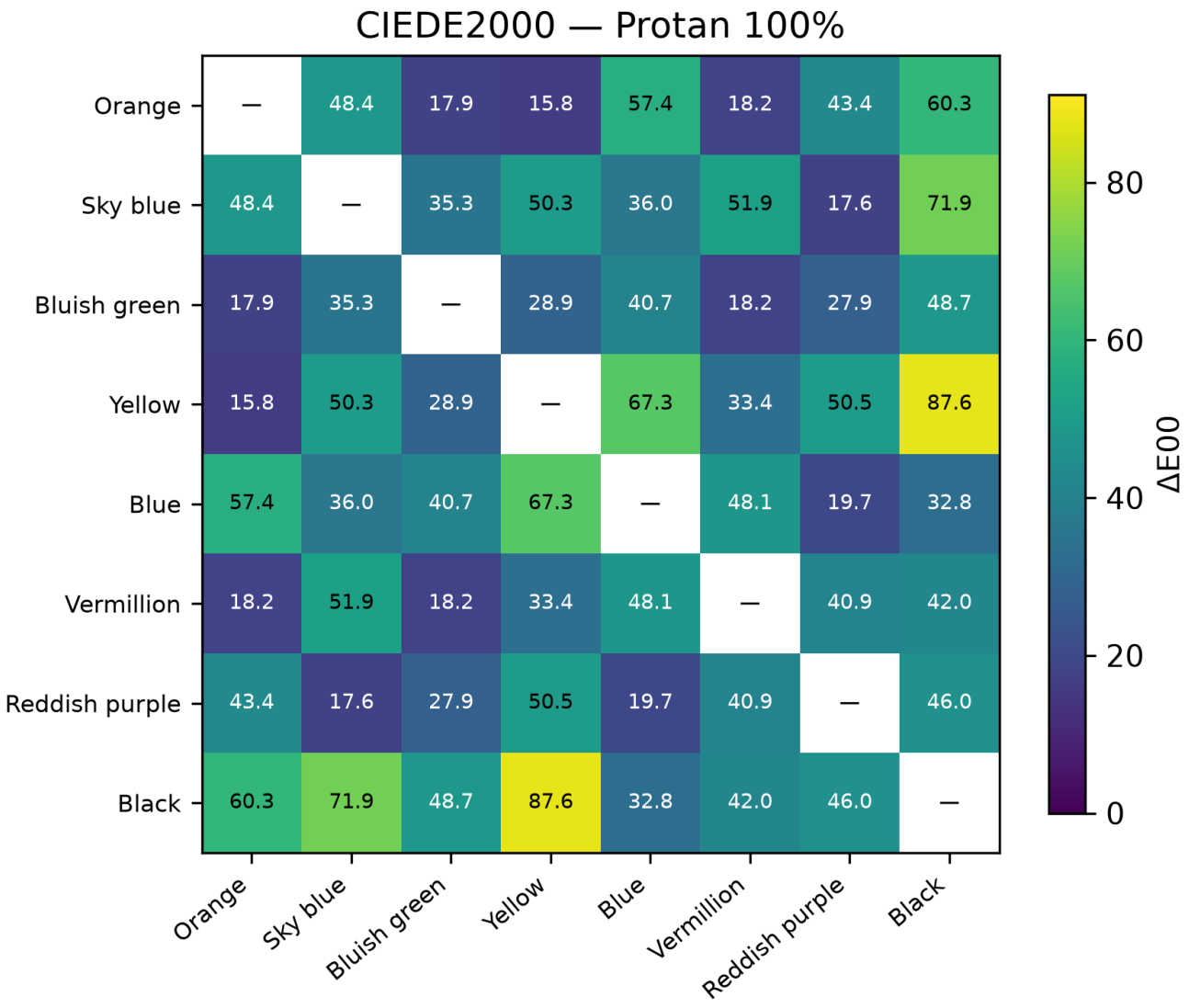
Figure 8. Minimum pairwise distance across simulated severity.

Normal pairwise heatmap



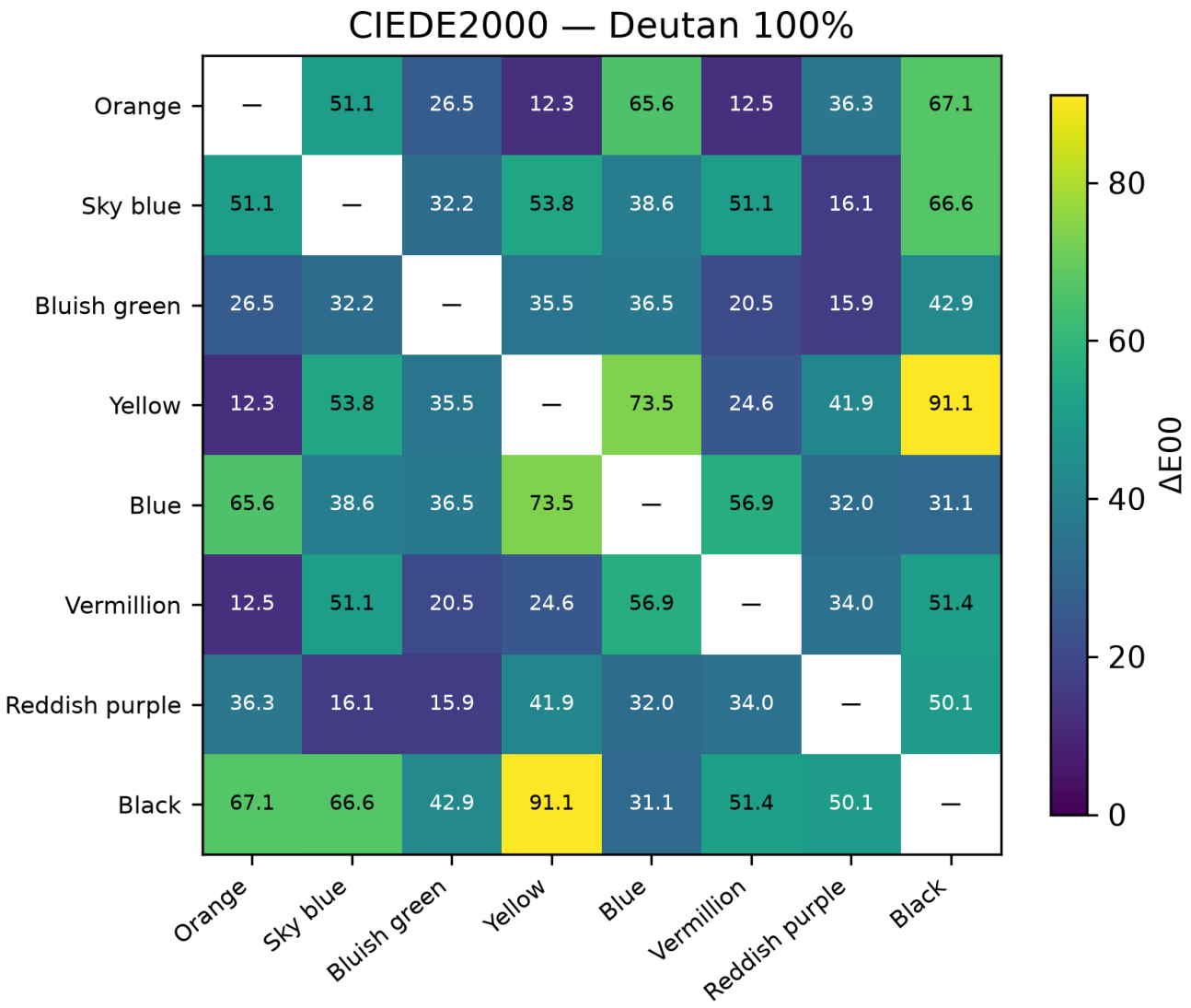
Normal pairwise heatmap.

Protan 100% pairwise heatmap



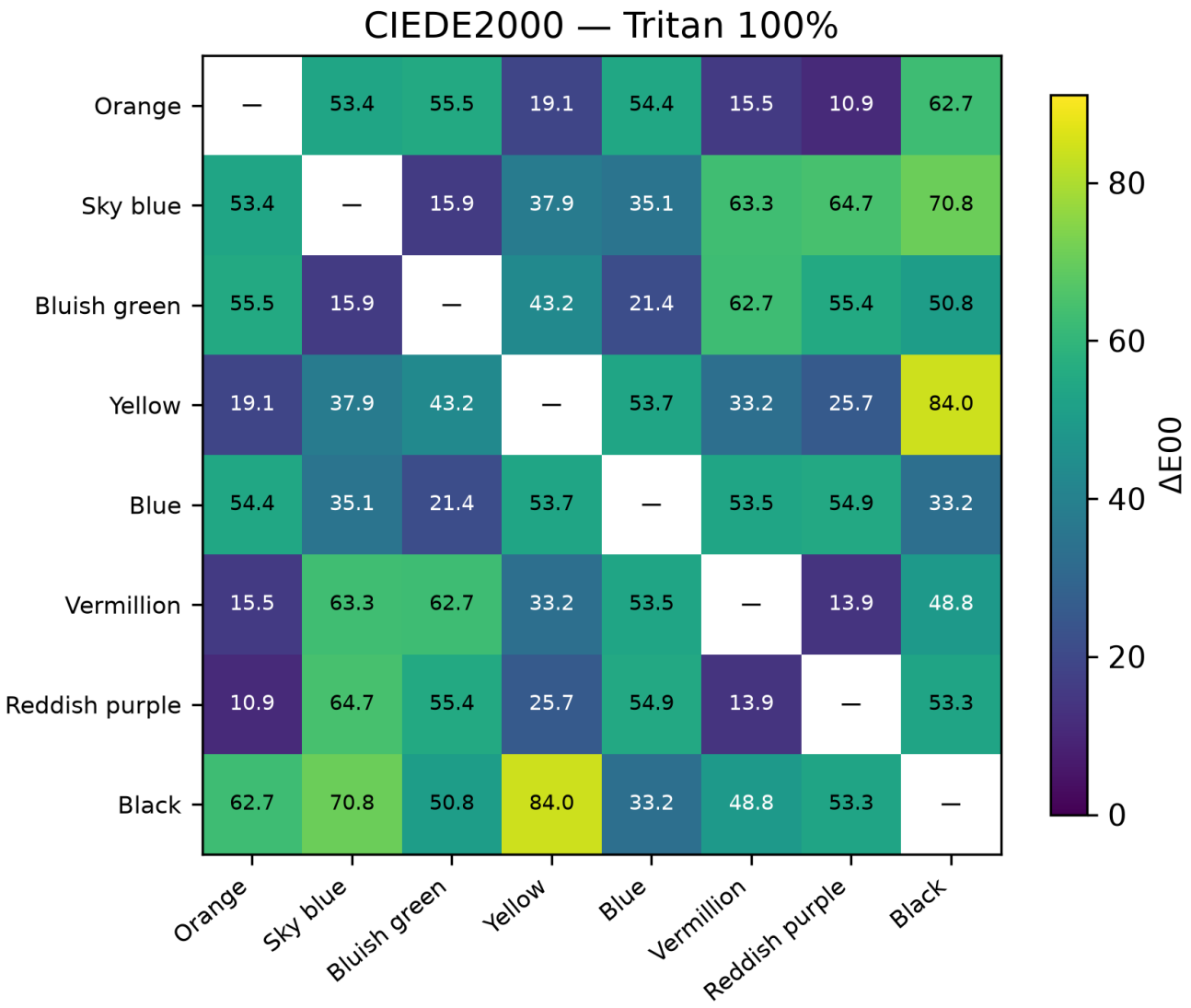
Protan 100% pairwise heatmap.

Deutan 100% pairwise heatmap



Deutan 100% pairwise heatmap.

Tritan 100% pairwise heatmap



Tritan 100% pairwise heatmap.

Normal diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Normal diagonal split pair matrix.

Protan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Protan 100% diagonal split pair matrix.

Deutan 100% diagonal split pair matrix

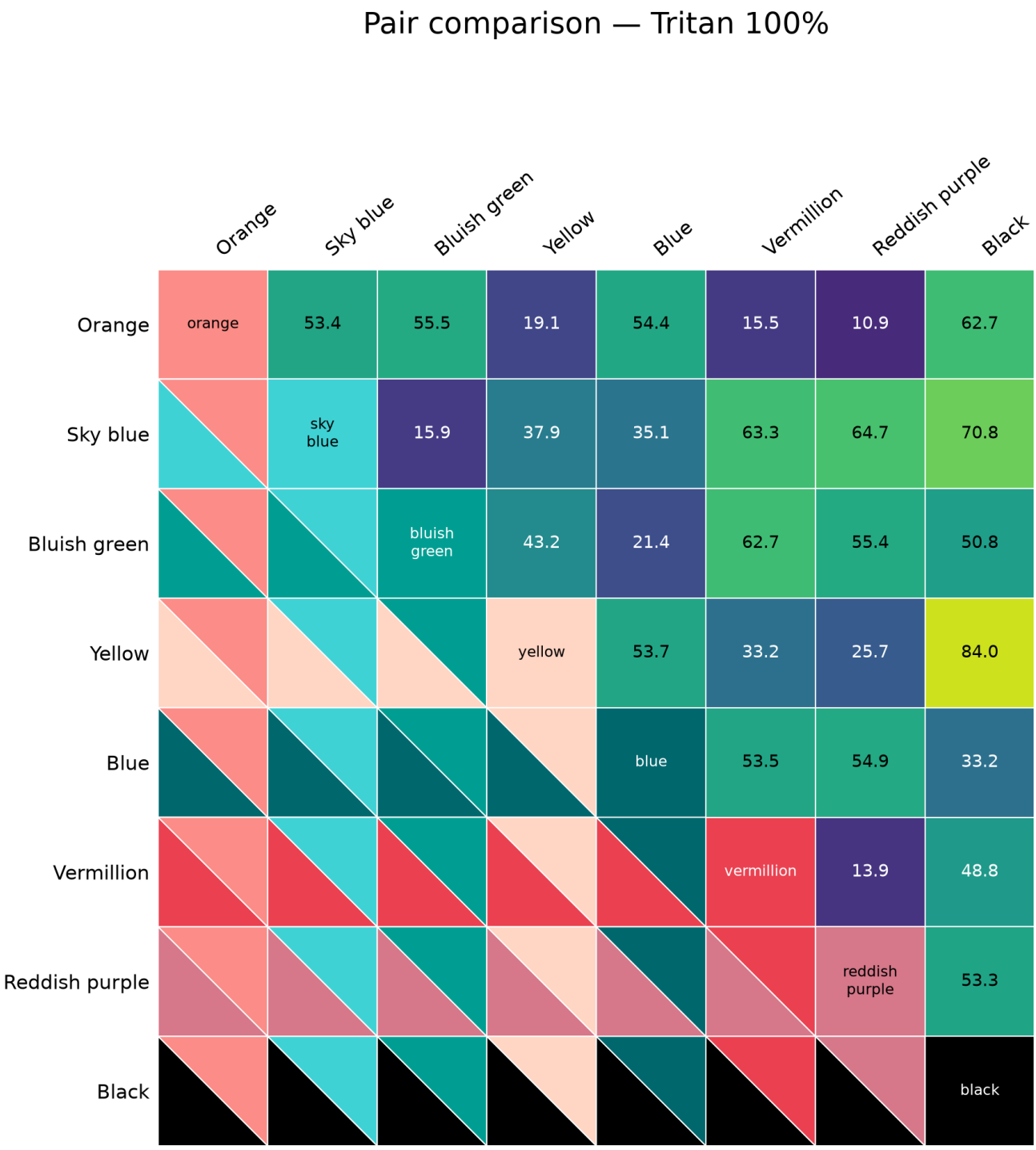
The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Deutan 100% diagonal split pair matrix.

Tritan 100% diagonal split pair matrix

The lower triangle places each colour pair in direct diagonal contact; the upper triangle reports its Delta E00 value.



Tritan 100% diagonal split pair matrix.

Grouped blue-family analysis

The YAML declares sky blue and blue as one visual family. With two members, the group contains one colour pair: sky blue and blue.

Condition	Severity	Pairs	Minimum	Mean	Maximum
normal	0	1	37.92	37.92	37.92
protan20	20	1	37.32	37.32	37.32
protan40	40	1	36.87	36.87	36.87
protan60	60	1	36.53	36.53	36.53
protan80	80	1	36.26	36.26	36.26
protan100	100	1	36.03	36.03	36.03
deutan20	20	1	37.83	37.83	37.83
deutan40	40	1	37.92	37.92	37.92
deutan60	60	1	38.10	38.10	38.10
deutan80	80	1	38.32	38.32	38.32
deutan100	100	1	38.57	38.57	38.57
tritan100	100	1	35.08	35.08	35.08
grayscale	0	1	37.43	37.43	37.43

Weakest-pair diagnostic



Figure 9. Weakest pairs under each standard condition.

Interpretation and limitations

The palette improves its intended blue-family separation in all standard PaletteBench conditions. It does not improve every pair. In particular, the darker blue approaches black, and its tritan-simulated relationship with vermillion loses separation relative to Okabe-Ito. Redundant encodings remain appropriate whenever category identity is critical.

CVD simulations and perceptual-distance models do not reproduce a particular person's perception. Monitor calibration, viewing environment, acuity, ageing, print reproduction, surrounding colours, spatial context, and mark size can all change discriminability. Human evaluation is outside this computational audit.

Reproducibility

Palette input SHA-256: b22d70f2358ec1174d3419bd880128eb929063add2013ee91c99f60b7cfaecf6. PaletteBench version: 0.1.0. Metric: CIEDE2000 via colour-science; sRGB D65 to CIE Lab. Simulation: Colorspacious sRGB1+CVD (Machado et al.) with display-bound clipping.

Responsible use of AI

This repository is the product of human-led research and development. The **conceptualisation, research objectives, scientific contributions, methodological choices, experimental design, interpretation of results, and conclusions** are determined by the human authors, who retain full intellectual and scientific responsibility for the work.

Generative artificial intelligence (AI) tools, such as ChatGPT and Codex, may be employed as **assistants** in software development and documentation. Their contributions may include implementing human-defined designs, suggesting or refactoring code, debugging, developing tests, improving documentation, exploring technical alternatives, and providing critical feedback. AI-generated suggestions are treated as provisional and are subject to human review, modification, and appropriate verification before adoption.

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Where AI tools are employed as part of the scientific methodology itself, rather than merely as development assistants, their role is documented separately as appropriate to ensure transparency and reproducibility.

References

1. Okabe, M., and Ito, K. (2002). How to make figures and presentations that are friendly to color blind people. Original brochure (PDF).

2. Wong, B. (2011). Points of view: Color blindness. *Nature Methods*, **8**(6), 441.

3. Wong, B. (2023). Author Correction: Points of view: Color blindness. *Nature Methods*, **20**(8), 1266.

4. Veen, R. J. (2026). PaletteBench (v0.1.0) [Software]. Zenodo. Source code for v0.1.0.